

Appendix A to Doc. 312 — The Closure Scale

Altermagnetism as a Laboratory Analogy:
Local versus Global Symmetry Breaking

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Appendix A to Doc. 312 — Altermagnetism as a Laboratory Analogy

Purpose of This Appendix

Doc. 312, § Lorentz (source lines 936–947), makes the statement:

On compact topology Lorentz invariance holds locally without restriction, but boosts are no longer global symmetries. The distinction is topological; no local measurement can detect it — only global markers.

The concept “locally broken, globally preserved” is abstract and difficult to ground without experimental reference. Altermagnetism provides a directly measured, mathematically fully analysed example of the same structure at a completely different scale. This appendix documents the analogy and marks its limits.

Purpose: Didactic reference and terminological calibration for “local $\neq$ global”. No new physical content for FFGFT. Source: Šmejkal, Sinova, Jungwirth, *Emerging Research Landscape of Altermagnetism*, Phys. Rev. X **12**, 040501 (2022).

.1 Altermagnetism: Three-Way Classification via Spin Groups [E]

The non-relativistic spin-group classification of collinear magnets yields exactly three phases, determined by which symmetry operation connects the magnetic sublattices:

Type	Sublattice link	Kramers deg.	M_{net}
Ferromagnet	none (one spin sublattice)	lifted (Zeeman)	$\neq 0$
Antiferromagnet	translation t or inversion P	preserved	$= 0$
Altermagnet	rotation C_n ($n = 2, 4, 6$)	locally lifted	$= 0$

The decisive distinction between antiferromagnet and altermagnet: in both, the net magnetisation is zero. But in the antiferromagnet a translation or inversion connects the sublattices — this *enforces* Kramers degeneracy $E(\mathbf{k}, \uparrow) = E(\mathbf{k}, \downarrow)$ across the entire Brillouin zone. In the altermagnet a genuine rotation (C_4 in RuO_2) connects the sublattices — this lifts the degeneracy in a direction-dependent manner.

.2 The Model Hamiltonian and Its Symmetries [E]

The effective single-particle Hamiltonian of the d_{xy} -wave altermagnet (Šmejkal et al., Eq. 3):

$$H(\mathbf{k}) = 2t \cos k_x \cos k_y \cdot \sigma_0 + 2t_J \sin k_x \sin k_y \cdot \sigma_z. \quad (1)$$

The energy eigenvalues are $E_{\pm}(\mathbf{k}) = 2t \cos k_x \cos k_y \pm 2t_J \sin k_x \sin k_y$ with spin splitting

$$\Delta E(\mathbf{k}) = E_+ - E_- = 4t_J \sin k_x \sin k_y. \quad (2)$$

Proof: C_4 maps the spin-up onto the spin-down Fermi surface [B]

The C_4 rotation in $\mathbf{k}$ -space acts as $(k_x, k_y) \rightarrow (-k_y, k_x)$. Then:

$$\begin{aligned} E_{\text{dn}}(-k_y, k_x) &= 2t \cos(-k_y) \cos(k_x) - 2t_J \sin(-k_y) \sin(k_x) \\ &= 2t \cos(k_y) \cos(k_x) + 2t_J \sin(k_y) \sin(k_x) = E_{\text{up}}(k_x, k_y). \quad \checkmark \end{aligned} \quad (3)$$

The spin-up Fermi surface is mapped exactly onto the spin-down surface by C_4 (numerically confirmed, `pruef_altermagnet_1_symmetrie.py`).

Two further analytically provable statements:

1. **Global compensation:** $\frac{1}{(2\pi)^2} \int_{\text{BZ}} \Delta E(\mathbf{k}) d^2k = 0$, since $\int_{-\pi}^{\pi} \sin k_x dk_x = 0$. No macroscopic net magnetisation. [B]
2. **Nodal lines:** $\Delta E = 0$ exactly at $k_x = 0$ or $k_y = 0$ (two orthogonal nodal lines, protected by mirror-plane symmetries that map one spin sublattice onto the other). [B]

.3 Locally Broken, Globally Preserved — the Shared Structure

Structural comparison with Doc. 312 § Lorentz

FFGFT / Doc. 312

Compact topology $T^4/\mathbb{Z}_3$
Lorentz invariance holds locally
Boosts: no longer global symmetries
Only topological markers detectable
Scale: Planck length

Altermagnet / Šmejkal 2022

Crystal lattice with C_4 rotation
 T -symmetry locally broken ($\Delta E \neq 0$)
Spin: globally compensated ($M = 0$)
Only ARPES/spin-current measurable
Scale: ångström (crystal lattice)

The structure is the same: a symmetry (time-reversal T or Lorentz boost) is locally broken in the sense that it fails at a single point (in $\mathbf{k}$ -space or for a specific direction of motion in space). But the global integral — over the entire Brillouin zone or over all directions — is zero: the overall structure is compensated. Locally this breaking is detectable only by direction-resolved measurements; an isotropic probe reveals nothing.

.4 Limits of the Analogy [E]

Restrictions — explicit

1. **Different scales.** Altermagnetism: $\sim$ Ångström. FFGFT: Planck scale ($\sim 10^{-35}$ m). No shared observables.
2. **Different symmetry groups.** The spin group $r \times R_s$ of the altermagnet and the orbifold group $T^4/\mathbb{Z}_3$ of FFGFT are different mathematical objects.
3. **No information flow.** FFGFT does not explain solid-state effects. Altermagnetism contributes no new derivations to FFGFT.

4. **Direction of the analogy.** The analogy serves the didactic calibration of the concept "locally $\neq$ globally". It is not a derivation and must not be read as a confirmation of FFGFT.

.5 Verification

All statements in §2 are numerically and symbolically confirmed by `pruef_altermagnet_1_symmetrie.py`. The script is located in `2/python/Dok312_Anhang_Skripte/`.

Reference: L. Šmejkal, J. Sinova, T. Jungwirth, *Emerging Research Landscape of Altermagnetism*, Phys. Rev. X **12**, 040501 (2022). arXiv:2204.10844.